M. ADITYA PATRA

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PROFESSIONAL SUMMARY

Data-driven B.Tech Computer Science student with hands-on experience building end-to-end data analysis and machine learning pipelines across scientific, healthcare, and database domains. Proficient in Python (Pandas, NumPy, Scikit-learn, XGBoost), SQL (MySQL), and data visualization (Matplotlib, Seaborn, Plotly). Delivered projects on datasets ranging from 300 to 9,500+ records, achieving up to 99% classification accuracy and ROC-AUC of 0.9989. Strong foundation in statistical hypothesis testing, feature engineering, and translating analytical findings into actionable insight. Seeking a Data Analyst or ML Engineer internship to drive decisions through rigorous, reproducible data science.

TECHNICAL SKILLS

Languages	Python, Java, C, SQL (MySQL)
Libraries	Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, XGBoost, Imbalanced-learn (SMOTE)
Tools & Platforms	Git, GitHub, Jupyter Notebook, VSCode, MySQL Workbench, Streamlit
Core Competencies	Exploratory Data Analysis, Statistical Hypothesis Testing (Welch t-test, Chi-square, ANOVA), Feature Engineering, Data Visualization, ML Classification, Model Evaluation (ROC-AUC, F1, Cross-Validation), SQL Query Design, Database Normalization
Methodologies	5-Fold Cross-Validation, Stratified Train-Test Split, SMOTE (Class Imbalance Handling), Scikit-learn Pipelines, Data Leakage Prevention

EXPERIENCE

Logistics & Organizing Member *2023 – Present*
Concordia Club, DY Patil University

- Coordinated logistics and participant management for 3+ large-scale university events including Model United Nations, handling 100+ participants across concurrent sessions
- Managed end-to-end event execution: scheduling, registration workflows, and on-ground coordination — developing organizational and cross-functional team skills applicable to deadline-driven professional environments

Data Analyst and Machine Learning Intern *June 2026 – August 2026*
CloudExperts.co

- Reviewed and analyzed production-grade codebases for a RAG web application built on React/TypeScript and AWS Lambda
- Gained hands-on exposure to Cloud Architecture and worked on Stock Market Analysis

EDUCATION

B.Tech in Computer Science & Engineering *2024-2028*
DY Patil International University, Pune

Class 12 *2023*
APS Bhopal

CERTIFICATIONS & TRAINING

- **Technology Commercialization** — University of Rochester (2024)
- **Data Mining in Python** — University of Michigan (2024)

PROJECTS

Kepler Exoplanet Classification *Apr 2026 – May 2026*
GitHub Repository 9,564 Records

- Built a machine learning pipeline that automates exoplanet candidate verification — a task that traditionally relies on expert astronomers manually reviewing telescope data — with near-perfect reliability (ROC-AUC 0.9989), demonstrating how ML can scale scientific discovery.
- Analyzed NASA Kepler mission dataset (9,564 exoplanet candidates, 50 orbital/stellar features) to distinguish confirmed planets from false positives using machine learning classification
- Addressed class imbalance (52.5% false positive, 24% confirmed) using SMOTE on the training set; evaluated models using F1 and ROC-AUC rather than accuracy to avoid misleading metrics on skewed classes
- Trained and compared three ensemble classifiers; Random Forest achieved **F1 0.9943 and ROC-AUC 0.9989** on a held-out test set of 1,241 objects — outperforming XGBoost (0.9981) and Gradient Boosting (0.9974)
- Proved statistical rigor by running Welch t-tests across all 12 key orbital and stellar features; every feature showed significant separation between classes (p ¡¡ ¡ 0.001), giving the model a scientifically justified — not just empirically lucky — foundation
- Deployed a Streamlit dashboard turning a static dataset into an interactive tool domain experts can use to explore candidates and get live predictions, bridging the gap between data science and astronomy research

Healthcare Data Analysis & Prediction *Feb 2026 – Mar 2026*
GitHub Repository 303 Clinical Records

- Translated raw clinical data into a risk-prediction tool and a set of statistically validated insights — the kind of analysis that helps clinicians prioritize which patients need further cardiac screening, rather than relying on intuition alone.
- Performed end-to-end analysis on the Cleveland Heart Disease dataset (303 patients, 13 clinical features) to identify statistically validated risk factors and build a clinical decision-support classifier
- Pinpointed exercise-induced angina, maximum heart rate, and chest pain type as the three strongest predictors of disease — actionable findings a clinician could use directly during patient triage
- Confirmed findings weren't due to chance: age ($p = 0.0001$) and maximum heart rate showed highly significant differences between patient groups; sex-disease association confirmed via Chi-square (p ¡ 0.0001)
- Built and compared Logistic Regression vs. Random Forest classifiers using a Scikit-learn Pipeline to prevent data leakage; Random Forest achieved **84% accuracy and ROC-AUC 0.909**, reliable enough to flag high-risk patients with few false negatives
- Validated the model would hold up on new patients, not just this dataset, via 5-fold cross-validation: Random Forest 83.8% ± 2.9% consistent performance across all folds
- Generated 6 visualizations specifically designed for clinical interpretability, translating model internals into insights a non-technical healthcare audience could act on

Football Archive Relational Database *Dec 2025 – Jan 2026*
GitHub Repository 9 Seasons, 3 Leagues

- Designed a production-grade relational database from scratch that mirrors how real sports analytics platforms (e.g., FBref, Opta) structure their data — proving the ability to architect, not just query, systems that scale across multiple leagues, seasons, and entity relationships.
- Designed and implemented a normalized MySQL database (3NF) with 11 tables spanning Premier League, La Liga, and Bundesliga match data across 3 seasons per league — covering matches, goals, players, transfers, standings, and season stats
- Enforced referential integrity using InnoDB engine with 16 foreign key constraints and STRICT SQL mode, eliminating the data inconsistency issues common in flat-file sports datasets
- Built 5 analytical views encapsulating complex multi-table JOIN logic, turning what would be repetitive, error-

prone manual queries into single, reusable lookups for match results, goal logs, live standings, and transfer history
- Developed 3 stored procedures for parameterized lookups (team fixtures, top-N scorers, head-to-head records) — the same query patterns a sports analytics dashboard or fantasy football app would need in production
- Wrote 12 analytical queries producing real football insights: xG over/under-performance per team, goals-per-90 contribution rankings, and clean sheet percentages — going beyond schema design to demonstrate actual domain-driven analysis
- Documented the full schema with an ER diagram and sample queries, showing the database is usable and extensible by others, not just a personal exercise

LANGUAGES

English (Fluent), Hindi (Fluent), Odia (Native), French (Basic)

Portfolio: https://portfolio-ztss.vercel.app/ | GitHub Stats: 8 public repositories with 50+ commits demonstrating consistent development practice